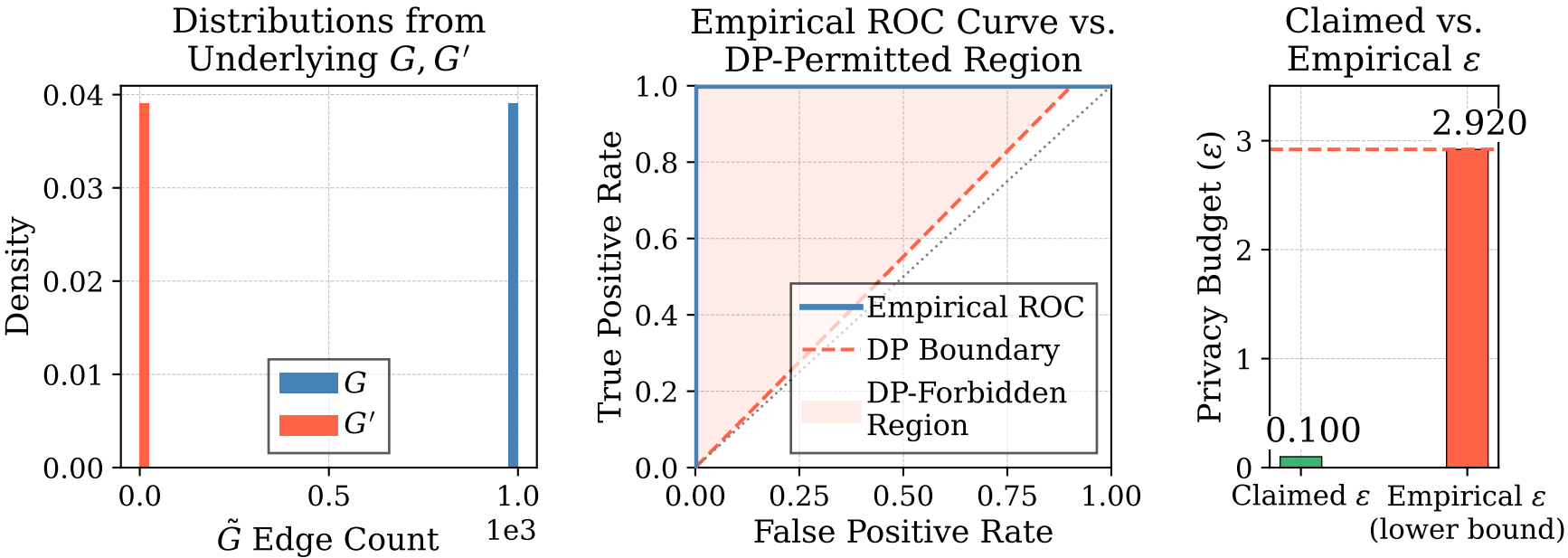
Construction-A (Directed) ($\varepsilon = 0.1$, $\delta = 1\text{e-}05$)
Distributions from Underlying G, G'
Density
0.04
0.03
0.02
0.01
0.00
G
G'
0.0
0.5
1.0
1e3
$\tilde{G}$ Edge Count
Empirical ROC Curve vs. DP-Permitted Region
True Positive Rate
1.0
0.8
0.6
0.4
0.2
0.0
Empirical ROC
DP Boundary
DP-Forbidden Region
0.00
0.25
0.50
0.75
1.00
False Positive Rate
Claimed vs. Empirical ε
Privacy Budget (ε)
3
2
1
0
2.920
0.100
Claimed ε
Empirical ε (lower bound)